Lecture 7

Braiding: hexagons, Yang–Baxter and topological spin

Objectives. *Lectures 5 and 6 built fusion categories, with their particles and associators. In three dimensions, exchanging two identical particles twice is the identity, and exchange statistics are those of the symmetric group: bosons and fermions. In a plane the two exchanges are not the same, and the symmetric group is replaced by the braid group. This lecture adds a braiding to a monoidal category, derives the braid relation from its axioms, writes the axioms as equations in F- and R-symbols, and solves them for Fibonacci. Closing a braided strand gives the topological spin.*

By the end of this lecture you will be able to:

- State the braid group and why it replaces the symmetric group in a plane.
- Define a braiding, prove the braid relation from the hexagons, and tell the braid form of the Yang–Baxter equation from *Integrability*'s form.
- Write the hexagon equations in F- and R-symbols and solve them for Fibonacci.
- Compute topological spins, and find the braidings of the pointed categories on $\mathbb{Z}_2$.

7.1 B_n versus S_n

The *braid group* B_n has generators $\sigma_1, \dots, \sigma_{n-1}$, with

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1},$$
$$\sigma_i\sigma_j = \sigma_j\sigma_i \quad (|i-j| \geq 2).$$

Adding $\sigma_i^2 = 1$ gives the symmetric group S_n. The generator σ_i exchanges particles i and $i+1$, passing the left one over:

$$\sigma_i = \quad \underset{1}{|} \cdots \underset{i}{\searrow} \underset{i+1}{\swarrow} \cdots \underset{n}{|}$$

Theorem 7.1 (Configuration spaces). *The fundamental group of the space of n unordered distinct points in $\mathbb{R}^2$ is B_n · cited (Kassel–Turaev §1.4.3); in $\mathbb{R}^d$ with $d \geq 3$ it is S_n · sketched.*

For $d \geq 3$: ordered configurations form $\mathbb{R}^{dn}$ with the diagonals $x_i = x_j$ removed, each of codimension $d \geq 3$, and removing subspaces of codimension at least 3 leaves a space simply connected. The unordered space is its quotient by the free action of S_n, so its fundamental group is S_n. In the plane the diagonals have codimension 2, like a point in a plane, and a loop taking one particle around another does not shrink. Quantum mechanics of identical particles in a plane therefore allows unitary representations of B_n, not only representations of S_n: *anyons* (*Symmetries* §15.4).

7.2 Braided categories and hexagons

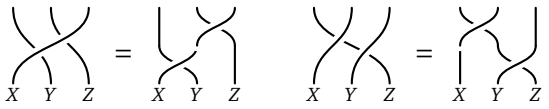
As in Lecture 4, we work as if the category were strict.

Definition 7.2 (Braiding)**.** A *braiding* on a monoidal category is a natural isomorphism $c_{X,Y} : X \otimes Y \to Y \otimes X$, drawn with the strand from bottom left to top right over the other, such that the *hexagons* hold:

$$\begin{aligned} &c_{X,Y\otimes Z} \\ &\quad = (\mathrm{id}_Y \otimes c_{X,Z}) \circ (c_{X,Y} \otimes \mathrm{id}_Z), \\ &c_{X\otimes Y,Z} \\ &\quad = (c_{X,Z} \otimes \mathrm{id}_Y) \circ (\mathrm{id}_X \otimes c_{Y,Z}). \end{aligned}$$

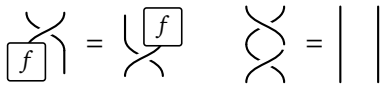
It is *symmetric* if $c_{Y,X} \circ c_{X,Y} = \mathrm{id}_{X\otimes Y}$ for all X, Y.

With the associators put back, each hexagon has six sides, hence the name (EGNO Def. 8.1.1). In pictures, the first says that a strand crossing over a pair crosses each in turn, and the second that a pair crossing over a strand does so one at a time:



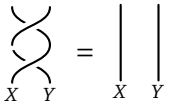
(7.1)

Naturality says that boxes slide through crossings, and invertibility that a crossing followed by its inverse is no crossing at all (Reidemeister II):



(7.2)

The *double braiding* $c_{Y,X} \circ c_{X,Y}$ takes one strand once around the other. A braiding is symmetric exactly when the double braiding is always trivial:



(7.3)

and then over- and under-crossings agree, and are drawn as a plain crossing.

Example 7.3 (Braidings)**.**

- Vect and $\mathrm{Rep}(G)$ with the flip $v \otimes w \mapsto w \otimes v$, which is an intertwiner and natural: symmetric.
- *Super vector spaces* sVec: Vec_G on $\mathbb{Z}_2 = \{1, s\}$ with $c_{s,s} = -1$ (EGNO Ex. 8.2.2). It is symmetric, since $(-1)^2 = 1$, and the odd line is a fermion: exchanging two of them gives a sign.
- The same category with $c_{s,s} = +1$: symmetric again, but a different braiding, not equivalent to the first as a braided category: an equivalence fixes s, so it would preserve $c_{s,s} = \pm 1$. A braiding is structure, not a property.

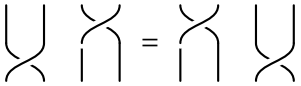
7.3 Yang–Baxter is Reidemeister III

Theorem 7.4 (The braid relation · *proved given coherence*)**.** *In a braided category, for all X, Y, Z,*

$$\begin{aligned}(c_{Y,Z}\otimes \mathrm{id}_X)(\mathrm{id}_Y\otimes c_{X,Z})&\\ \circ (c_{X,Y}\otimes \mathrm{id}_Z)&\\ = (\mathrm{id}_Z\otimes c_{X,Y})(c_{X,Z}\otimes \mathrm{id}_Y)&\\ \circ (\mathrm{id}_X\otimes c_{Y,Z}).&\end{aligned}$$

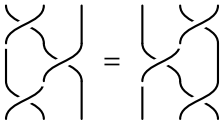
So $\sigma_i \mapsto \mathrm{id}\otimes c_{X,X}\otimes \mathrm{id}$, in positions $i, i+1$, is a representation of B_n on $X^{\otimes n}$.

Proof. By the first hexagon the last two factors on the left are $c_{X,Y\otimes Z}$. Naturality of $c_{X,-}$ applied to $c_{Y,Z}\colon Y\otimes Z\to Z\otimes Y$ gives $(c_{Y,Z}\otimes \mathrm{id}_X)\,c_{X,Y\otimes Z} = c_{X,Z\otimes Y}\,(\mathrm{id}_X\otimes c_{Y,Z})$, and the first hexagon again turns $c_{X,Z\otimes Y}$ into the first two factors on the right. The far commutation $\sigma_i\sigma_j = \sigma_j\sigma_i$ is the interchange law (Theorem 3.4):



■

In pictures this is Reidemeister III: a strand slides over a crossing.



(7.4)

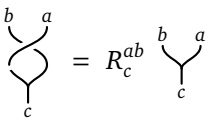
Let P be the flip, $P(v\otimes w) = w\otimes v$. For a map R on $V\otimes V$, a leg subscript specifies which ordered factors of $V^{\otimes 3}$ it acts on, leaving the other fixed:

$$R_{12} = R\otimes \mathrm{id}, \qquad R_{23} = \mathrm{id}\otimes R,$$
$$R_{13} = P_{23}(R\otimes \mathrm{id})P_{23}.$$

Integrability's Yang–Baxter equation carries spectral parameters (its eq. (14.8)). For a constant matrix R on $V\otimes V$ it reads $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, and then R satisfies it exactly when $c = P\circ R$ satisfies the braid form $c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}$, which Kassel (Thm. XIII.1.3) and EGNO (Prop. 8.1.10) call the Yang–Baxter equation. Three things are called R, and must be kept apart: *Integrability*'s $R(\lambda)$, the braiding $c = P\circ R$, and the R-symbols of the next section.

7.4 R-symbols; Fibonacci solved

In a multiplicity-free fusion category with the vertices of Lecture 5, the braiding maps the line $\mathrm{Hom}(c, a\otimes b)$ to the line $\mathrm{Hom}(c, b\otimes a)$, so it acts on vertices by numbers, the *R-symbols*:



(7.5)

that is $c_{a,b}\circ \psi_c^{ab} = R_c^{ab}\,\psi_c^{ba}$, with $R_c^{ab} = 1$ when a or b is 1: the first hexagon with $Y = Z = 1$ gives $c_{X,1} = c_{X,1}^2$, so $c_{X,1} = \mathrm{id}$, and likewise $c_{1,X} = \mathrm{id}$. In a unitary category $\left|R_c^{ab}\right| = 1$. A gauge transformation (Theorem 6.4) changes R_c^{ab} by u_c^{ba}/u_c^{ab}, so R_c^{aa} is a gauge invariant.

Theorem 7.5 (The hexagon equations · *proved given coherence*). *The F- and R-symbols of a braided multiplicity-free fusion category satisfy, for all labels,*

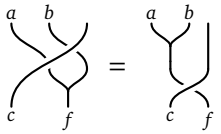
$$R_e^{ca}\,[F_d^{acb}]_{eg}\,R_g^{cb} = \sum_f [F_d^{cab}]_{ef}\,R_d^{cf}\,[F_d^{abc}]_{fg},$$

and, from the second hexagon,

$$(R_e^{ac})^{-1}[F_d^{acb}]_{eg}(R_g^{bc})^{-1} = \sum_f [F_d^{cab}]_{ef}\,(R_d^{fc})^{-1}[F_d^{abc}]_{fg}$$

(Bonderson (2.57)–(2.58)). Conversely, F- and R-symbols satisfying the pentagon and both families define a braiding · *sketched.*

Proof. The step that does the work is that a strand crosses a vertex as it crosses the two strands leaving it, by naturality and the first hexagon:



(7.6)

Apply both sides of the first hexagon, with $X = c$, $Y = a$, $Z = b$, to the tree in which a and c fuse first, through e, to total charge d. On the right, c crosses a, which multiplies by R_e^{ca}; an F-move brings c and b together, through g; c crosses b, which gives R_g^{cb}. On the left, an F-move brings a, b together, through f; c crosses the pair f at once, giving R_d^{cf}; a last F-move returns to the basis of trees fusing b and c first. Comparing coefficients of the basis trees gives the equation. The second family comes the same way from the second hexagon, written for the inverse braiding: $c_{b,a}^{-1}$ acts on ψ_c^{ab} by $(R_c^{ba})^{-1}$, so inverting a crossing also exchanges its labels. The converse is the argument of Theorem 6.3. ∎

For Fibonacci, with F from Theorem 6.6, the unknowns are $R_1 = R_1^{\tau\tau}$ and $R_\tau = R_\tau^{\tau\tau}$. Two instances of the first family, with $a = b = c = \tau$, are enough. With $d = 1$ and $e = g = \tau$, every F-symbol is 1 and the sum has only $f = \tau$:

$$R_\tau^2 = R_1.$$

With every label τ, the left side is $R_\tau(-\varphi^{-1})R_\tau$ and the right side $\varphi^{-1/2}\cdot 1\cdot\varphi^{-1/2} + (-\varphi^{-1})R_\tau(-\varphi^{-1})$, so

$$R_\tau^2 + \varphi^{-1}R_\tau + 1 = 0.$$

Theorem 7.6 (Fibonacci's R-symbols). *The hexagons for Fibonacci have exactly two solutions,*

$$R_1^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5},$$

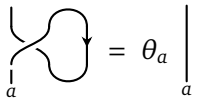
and its complex conjugate.

Proof. The roots of $x^2 + \varphi^{-1}x + 1 = 0$ have product 1 and sum $-\varphi^{-1} = 2\cos(3\pi/5)$, so they are $e^{\pm 3\pi i/5}$. Then $R_1 = R_\tau^2 = e^{\pm 6\pi i/5} = e^{\mp 4\pi i/5}$. Both pairs satisfy every instance of both families (`scripts/check-l7-algebra.py`). ∎

Bonderson (p. 82) and Simon (eq. (10.2)) print the first. The picture alone does not choose; the spin of the next section does. The resulting action of B_3 on three τ's is Sheet 7.

7.5 Topological spin

Closing the right strand of $c_{a,a}$, its partial trace in the sense of (4.9), gives an endomorphism of the simple a, hence a number, the *topological spin* θ_a:



(7.7)

Theorem 7.7 (The spin from R · *proved given coherence*)**.** *In a unitary multiplicity-free fusion category with its canonical spherical structure,*

$$\theta_a = \frac{1}{d_a}\sum_c N_{aa}^{c}\, d_c\, R_c^{aa}.$$

Proof. Take the trace of both sides of (7.7): the left side becomes the trace of $c_{a,a}$, and the right side $\theta_a d_a$. On each channel c of $a \otimes a$, $c_{a,a}$ acts as R_c^{aa} by (7.5), so it is $\sum_c R_c^{aa} P_c$ with P_c the projections of (5.3), and the trace of P_c is $N_{aa}^{c} d_c$. ∎

For Fibonacci,

$$\theta_\tau = \frac{e^{-4\pi i/5} + \varphi\, e^{3\pi i/5}}{\varphi} = e^{4\pi i/5},$$

and the conjugate solution gives $e^{-4\pi i/5}$. The course takes $\theta_\tau = e^{4\pi i/5}$, which fixes the chirality. The spin is a gauge invariant, like the double braiding: a gauge changes R_c^{ab} by u_c^{ba}/u_c^{ab} and R_c^{ba} by the inverse. The two are related by the *ribbon relation* $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$ (Bonderson (2.60) · *cited*); Lecture 9 identifies θ_a with the twist of a ribbon category, the phase of rotating one particle by 2π.

7.6 Pointed braidings; Ising

For Vec_G^{ω} on $\mathbb{Z}_2 = \{1, s\}$, a braiding is determined by one number, written $c(s,s) = R_1^{ss}$. By Theorem 7.7, $\theta_s = c(s,s)$.

Theorem 7.8 (Braidings on $\mathbb{Z}_2$)**.** *The hexagons for* Vec_G^{ω} *on* $\mathbb{Z}_2$ *force* $c(s,s)^2 = \omega(s,s,s)$*, and each such* $c(s,s)$ *gives a braiding* · *sketched. So the trivial* ω *has two braidings,* $c(s,s) = \pm 1$*, the* boson *and the* fermion *(*sVec*); and* $\omega(s,s,s) = -1$ *has two,* $c(s,s) = \pm i$*, the* semion *and the* antisemion*.*

Proof. In the first family take $a = b = c = d = s$ and $e = g = 1$, so that $f = 1$: the left side is $c(s,s)\,\omega(s,s,s)\,c(s,s)$ and the right side $\omega(s,s,s)\cdot 1\cdot\omega(s,s,s)$. So $c(s,s)^2 = \omega(s,s,s)$. Conversely every instance of both families holds when it does (`scripts/check-l7-algebra.py`), and the converse of Theorem 7.5 gives the braiding. ∎

Here ω was fixed and the braiding varied. A *quadratic form* on a finite abelian group A is a function $q\colon A \to \mathbb{C}^\times$ with

$$q(-x) = q(x) \qquad (x \in A),$$

for which the ratio

$$\frac{q(x+y)}{q(x)q(y)} \qquad (x, y \in A)$$

is multiplicative in each argument. Pointed braided fusion categories are classified, up to braided equivalence, by these forms up to automorphisms of A · *cited*. Every form occurs,

with $q(x) = \theta_x$, and determines ω up to a coboundary. For $\mathbb{Z}_5$, the five possible functions $q(x) = e^{2\pi i k x^2/5}$ give three classes: $k = 0$, $k \in \{1, 4\}$ and $k \in \{2, 3\}$ (EGNO Thms. 8.4.9 and 8.4.12, after Eilenberg–Mac Lane and Joyal–Street).

Theorem 7.9 (Braidings on Ising · *cited*)**.** *Each of the two Ising fusion categories has exactly four braidings (Siehler, Thm. 1.2). On the course's Ising ($\kappa = +1$) they have $\theta_\sigma = e^{i\pi\nu/8}$ with $\nu \in \{1, 7, 9, 15\}$; the other odd ν belong to the $\kappa = -1$ category, which contains $\mathrm{SU}(2)_2$ ($\nu = 3$), and $\kappa_\sigma = (-1)^{(\nu^2-1)/8}$ (Kitaev, eqs. (91)–(96) and Table 1).*

The course's Ising is $\nu = 1$; in Lecture 6's gauge (Kitaev's Table 1, Bonderson p. 84),

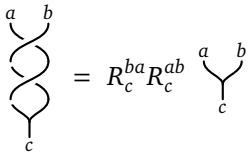
$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{3i\pi/8},$$
$$R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma} = -i, \quad R_1^{\psi\psi} = -1.$$

With the F of Lecture 6 these satisfy both hexagon families (the script checks them; Sheet 7 solves them). The spin formula gives

$$\theta_\sigma = \frac{e^{-i\pi/8} + e^{3i\pi/8}}{\sqrt{2}} = e^{i\pi/8}\,\frac{2\cos(\pi/4)}{\sqrt{2}} = e^{i\pi/8}.$$

The spin of ψ is $\theta_\psi = -1$, so ψ is a fermion.

On a vertex the double braiding is a number, since both crossings act by R-symbols:



$$= R_c^{ba} R_c^{ab} \tag{7.8}$$

For the toric code's e and m it is -1, the phase *Quantum Computing* measured by taking e around m (Sheet 7). Finally, an object X that braids trivially with everything, $c_{Y,X}c_{X,Y} = \mathrm{id}$ for all Y, is *transparent*; the transparent objects form the *Müger centre* (EGNO's centralizer C', §8.20). In sVec every object is transparent. In Fibonacci the only transparent simple object is 1, since $R_c^{\tau\tau}R_c^{\tau\tau} \neq 1$ for both channels. A *trivial* Müger centre consists of direct sums of the unit. Lecture 10 shows that this holds exactly when the S-matrix is invertible.

Summary.

- *Handed on:* in a plane, exchanges form the braid group B_n (Theorem 7.1); a braiding is a natural $c_{X,Y}$ with two hexagons (Definition 7.2), and it gives a representation of B_n (Theorem 7.4); Yang–Baxter in braid form is Reidemeister III.
- *R-symbols:* $c_{a,b}\psi_c^{ab} = R_c^{ab}\psi_c^{ba}$, constrained by the hexagon equations in F and R (Theorem 7.5); for Fibonacci $R_1^{\tau\tau} = e^{-4\pi i/5}$, $R_\tau^{\tau\tau} = e^{3\pi i/5}$ or the conjugate (Theorem 7.6).
- *Spin:* $\theta_a = d_a^{-1}\sum_c N_{aa}^c d_c R_c^{aa}$, the value of the closed curl (Theorem 7.7); $\theta_\tau = e^{4\pi i/5}$.
- *Pointed and Ising:* on $\mathbb{Z}_2$, $c(s,s)^2 = \omega(s,s,s)$: boson, fermion, semion, antisemion (Theorem 7.8); Ising has four braidings, and the course's has $\theta_\sigma = e^{i\pi/8}$ (Theorem 7.9).
- *Next:* Lecture 8 builds braidings from Hopf algebras: quantum groups.

References

Kassel and Turaev, *Braid Groups*: B_n as a fundamental group, §1.4.3, p. 29. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: braidings and hexagons, Def. 8.1.1, p. 195; the Yang–Baxter equation, Prop. 8.1.10, p. 196; symmetric categories and sVec, Def. 8.1.12 and Ex. 8.2.2, p. 197; pointed braidings, Ex. 8.4.3, p. 204, and quadratic forms, Thms. 8.4.9 and 8.4.12, pp. 205–206; the centralizer, §8.20, p. 239. Kassel, *Quantum Groups*: braidings, Def. XIII.1.1, p. 315, and the Yang–Baxter equation, Thm. XIII.1.3, p. 317. Heunen and Vicary, *Categories for Quantum Theory*, Def. 1.17, p. 40. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438: Ising, eqs. (91)–(96), p. 40, and Table 1, p. 30; braiding, eqs. (204)–(209), pp. 80–81; spin, eqs. (211)–(215), p. 82. Bonderson, PhD thesis (Caltech, 2007): R-moves and hexagons, (2.53)–(2.58), pp. 25–26; the ribbon relation and spin, (2.60)–(2.61), p. 27; Fibonacci and Ising, pp. 82–84. Simon, *Topological Quantum*: braid groups, §3.3, pp. 28–31; Fibonacci, eq. (10.2), p. 126; hexagons, §13.3, p. 173. Siehler, *Braided near-group categories*, math/0011037, Thm. 1.2.